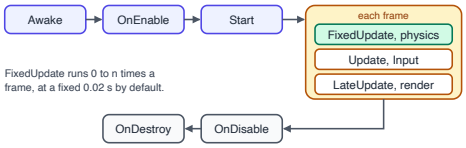
Unity Cheat Sheet

Unity 6.3 LTS (6000.3) · URP · changes up to 6.6 · checked 5 November 2026 · Irurueta Docs, Graphics & Games

Versions and plans (November 2026)

Item	Status
LTS	**6.3** until Dec 2027; 6.0 through Oct 2026
Supported	**6.6**, the last of the Unity 6 family
Next	6.7 LTS beta, due end of 2026; 7.0 public alpha, not for production
Hub	3.22.5 (28 Oct 2026)
Personal	Free, up to US$200k revenue + funding; splash optional
Pro	US$2,310 per seat per year; required above US$200k
Enterprise	Custom; required above US$25M
Runtime Fee	Cancelled 12 Sep 2024

MonoBehaviour lifecycle



Core API one-liners

```
var rb = GetComponent<Rigidbody>();
TryGetComponent(out Collider col);
var b = Instantiate(prefab, pos, rot);
Destroy(b, 2f);
float dt = Time.deltaTime;
transform.Rotate(0, speed * dt, 0);
transform.position += dir * (speed * dt);
var hud = FindAnyObjectByType<Hud>();
StartCoroutine(Fade());
IEnumerator Fade() {
  yield return new WaitForSeconds(1f);
}
async Awaitable FadeAsync() {
  await Awaitable.WaitForSecondsAsync(
    1f, destroyCancellationToken);
}
```

Input System

```
InputAction move, grab;
void Awake() {
  var map = InputSystem.actions;
  move = map.FindAction("Player/Move");
  grab = map.FindAction("Player/Grab");
  grab.performed += _ => Grab();
}
void Update() {
  Vector2 v = move.ReadValue<Vector2>();
}
```

Polling in `Update`, callbacks for events. Unsubscribe in `OnDestroy`.

Physics

- **Rule:** forces in `FixedUpdate`, input reads in `Update`
- `rb.linearVelocity`, `rb.linearDamping`, `rb.AddForce(f, ForceMode.Impulse)`
- Interpolate on the Rigidbody the camera follows

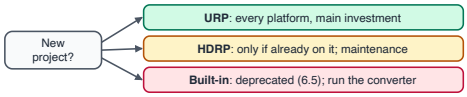
```
if (Physics.Raycast(
    ray, out var hit, 100f, mask))
  Select(hit.collider);
```

Colliders	Callback
Solid, one with Rigidbody	`OnCollisionEnter`
Is Trigger on either	`OnTriggerEnter`

Serialisation attributes

Attribute	Effect
`[SerializeField]`	private field in the Inspector
`[HideInInspector]`	saved, not shown
`[System.NonSerialized]`	not saved
`[Header]`, `[Tooltip]`, `[Range(0, 1)]`	Inspector layout
`[SerializeReference]`	polymorphic references
`[FormerlySerializedAs]`	rename, keep data
`[CreateAssetMenu]`	ScriptableObject menu item
`[RequireComponent]`	adds a dependency

Render-pipeline decision



Render Graph is mandatory for URP custom passes since 6.4.

Lighting and textures

Light mode	Use
Baked	static shelf; lightmaps, APV
Mixed	baked bounce, real-time direct
Realtime	moving lamp; costs per frame
Platform	**Texture format**
Android, iOS	ASTC (PVRTC removed in 6.4)
Windows, macOS, Linux	BC7, DXT5
Web	ASTC or DXT, by device

GPU lightmapper by default; the CPU baker is deprecated (6.6). Use Linear colour space.

UI Toolkit or uGUI

	UI Toolkit	uGUI
Model	UXML, USS, C#	GameObjects, Canvas
Text	built in	TextMeshPro in `ugui` 2.0
Use for	new screens, Editor UI, data binding	existing UI, custom VFX in UI

Animation

```
animator.SetFloat("Speed", v.magnitude);
animator.SetTrigger("Grab");
```

- Hash names with `Animator.StringToHash`
- **Cinemachine 3:** `CinemachineCamera`, one Brain on the Camera
- **Timeline:** PlayableDirector; signals call methods

Patterns map

Need	Pattern
Game flow	State
Undo, replay	Command
Decoupled HUD	Observer, event channel, Event Bus
Many short-lived objects	Object Pool (`UnityEngine.Pool`)
Swappable rules	Strategy, ScriptableObject
Global service	Service Locator; avoid Singleton
Near-object queries	Spatial Partition
Testable UI	MVP, MVVM

Minimal DOTS system

```
[BurstCompile]
public partial struct SpinSystem
    : ISystem {
  [BurstCompile]
  public void OnUpdate(
      ref SystemState state) {
    float dt = SystemAPI.Time.DeltaTime;
    foreach (var (t, s) in SystemAPI.Query<
        RefRW<LocalTransform>,
        RefRO<Spin>>())
      t.ValueRW = t.ValueRO
        .RotateY(s.ValueRO.Speed * dt);
  }
}
```

Builds and platforms

- **Build Profiles:** File > Build Profiles, one per target
- **IL2CPP:** required on iOS and Web, for release on Android; Mono builds faster for iteration
- **Web:** WebGL 2 by default; WebGPU production from 6.6; Wasm 2023 default from 6.5; serve Brotli or gzip with `Content-Encoding`
- **Android:** API 26 (6.5), OpenGL ES 3.1 (6.6)

```
#if UNITY_ANDROID || UNITY_IOS
#elif UNITY_WEBGL
#elif UNITY_EDITOR
#endif
```

Profiling checklist

- Profile a development build on the target device
- Budget: 16.6 ms at 60 fps, 33.3 ms at 30 fps
- CPU or GPU bound? Profiler, Frame Debugger
- GC Alloc in `Update` should read 0
- SRP Batcher, GPU Resident Drawer, pooling
- Memory Profiler, Project Auditor, markers

Migration from 2020 and 2021 tutorials

Old	Now
`Input.GetAxis`	Input System actions
`rb.velocity`, `drag`	`linearVelocity`, `linearDamping`
`FindObjectOfType`	`FindAnyObjectByType`
`OnGUI` HUD	UI Toolkit
Standard shader	URP Lit
Build Settings	Build Profiles
WebGL platform	Web
`BinaryFormatter`	JSON
`go.rigidbody`	`GetComponent` (6.5)
Statics reset on Play	Reset them yourself (6.6)
`InstanceID`	`EntityId`, 64-bit

Editor shortcuts (Cmd on macOS)

`Q W E R T`	hand, move, rotate, scale, rect
`F`	frame the selection
`Ctrl+P`, `Ctrl+Shift+P`	play, pause
`Ctrl+D`	duplicate
`Ctrl+S`	save the scene
`Ctrl+Shift+N`	new empty GameObject
`Ctrl+B`	build and run
`Alt` + drag	orbit the Scene view